

37 Analytical techniques

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A Level spectra may combine high-resolution mass spectrometry, IR, $^1\text{H}/^{13}\text{C}$ NMR and chromatography. Use molecular formula/degree of unsaturation first, then identify carbonyl and O-H signals and proton environments; no single peak proves an entire structure. Chromatographic R_f = distance spot/distance solvent front for the stated stationary and mobile phases.

Worked example: follow the reasoning

If a chromatogram spot travels 3.2 cm and solvent front 8.0 cm, R_f = 0.40 under those conditions.

Quick check · multiple choice

A An M and M+2 mass peak in near 3:1 ratio suggests which atom? [1]

- A Chlorine
- B Carbon only
- C Nitrogen
- D Hydrogen

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A mass spectrum suggests formula $\text{C}_3\text{H}_6\text{O}$ and IR has strong $\text{C}=\text{O}$; ^1H NMR has one proton environment. Suggest structure and justify. [3]

Answers and explanation

Quick check: A. Chlorine

Question 1 [3 marks]

Propanone CH_3COCH_3 [1]; equivalent methyl groups give one ^1H environment [1]; carbonyl accounts for IR and formula [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.